

CivicMitra

Technology Stack

Comprehensive Technical Specification for Smart Civic Governance

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Target: Smart India Hackathon & Production Deployment

1 Frontend (Citizen + Admin UI)

Layer	Technology	Why we chose it
Citizen App	React + Vite + Tailwind CSS (PWA)	Fast development, mobile-first, offline support, installable like an app without App Store, responsive for all devices.
Admin Dashboard	React + Tailwind + React Query / Zustand	Quick, responsive, role-based dashboard for municipal staff, easy integration with maps and charts.
Maps / Geo	Leaflet.js or Mapbox	Lightweight, interactive map support, can render pins, heatmaps, and ward boundaries. Works offline if tiles cached.
Notifications	Firebase Cloud Messaging / Twilio SMS	Push notifications and SMS for status updates, works even in rural low-connectivity areas.

Reasoning:

React + PWA allows users to report issues with minimal friction, even on low-end devices. Tailwind makes UI consistent and fast to develop. Map libraries are lightweight and flexible. SMS fallback ensures rural reach.

2 Backend (APIs, Routing, Auth, Analytics)

Layer	Technology	Why we chose it
API Server	Node.js + Express / NestJS	Lightweight, scalable, easy to implement routing logic, connects seamlessly with geospatial DB and storage.
Database	PostgreSQL + PostGIS	Relational DB for structured reports, PostGIS handles geospatial queries (auto-routing, ward/panchayat mapping).
Media Storage	AWS S3 / Firebase Storage / Google Cloud Storage	Secure storage for images, voice notes; CDN support for fast media delivery.
Auth	Firebase Auth / Keycloak / Autho	Secure, role-based access control; supports OTP/2FA; easy integration with mobile/web apps.
Analytics	PostgreSQL + Grafana / Metabase / Superset	Generate dashboards, heatmaps, KPIs; track SLA compliance and citizen engagement.

Reasoning:

Node.js + PostgreSQL is scalable and maintainable. PostGIS ensures auto-routing based on GPS. Cloud storage handles large multimedia uploads. Firebase/Autho provides secure login and user management quickly.

3 Geospatial & Routing

Ward / Panchayat boundaries database for urban/rural detection.

Backend logic: Category + Geo → Correct Department → Field Officer.

Frontend visualization: Leaflet/Mapbox for live issue map, heatmaps.

Reasoning:

Citizens report issues without choosing department. Auto-routing via geospatial data ensures accuracy and simplicity.

4 Security & Compliance

Encryption: TLS 1.3 + HSTS, secure cookies

Access Control: RBAC (Reporter, Field Officer, Supervisor, Admin, Auditor)

Data Protection: Encryption at rest (AES-256) + in transit (TLS 1.3)

Audit Logging: Append-only immutable logs

Data Residency: Inside India/state cloud (NIC / AWS Gov / Azure India)

Reasoning:

Security and privacy are critical for government apps. Compliance with Indian regulations ensures trust and smooth deployment.

5 Deployment / Hosting

Option	Use Case
Hackathon / MVP	Firebase Hosting + Functions or Vercel / Netlify + Managed DB
Pilot / State rollout	Government cloud (NIC / AWS Gov / Azure India)
Offline / Rural support	PWA caching, service workers, background sync

Reasoning:

Start fast for hackathon. Later, migrate to fully compliant cloud for production and scale.

6 Future-ready / ML Integration

Auto-priority scoring, duplicate detection, predictive maintenance.

Image classification (potholes, streetlight types).

API designed modularly so ML can be added later without major refactor.

Reasoning:

Allows MVP to be simple and stable, while leaving room for AI/ML enhancements after pilot.

Technology Stack Summary

Layer	Technology	Key Advantage
Frontend (Citizen)	React + PWA + Tailwind	Fast, mobile-first, offline, low friction
Frontend (Admin)	React + Tailwind + React Query	Responsive dashboards, map & analytics
Maps / Geo	Leaflet / Mapbox	Interactive map + heatmaps, geo-routing
Backend API	Node.js + Express/NestJS	Scalable, modular, fast integration
Database	PostgreSQL + PostGIS	Relational + geo queries for auto-routing
Media Storage	AWS S3 / Firebase Storage	Secure, CDN-enabled, scalable media storage
Auth	Firebase Auth / Keycloak	Secure, role-based, OTP/2FA support
Notifications	Firebase Cloud Messaging / Twilio SMS	Push & SMS notifications for rural users
Analytics	Grafana / Metabase / Superset	KPI dashboards, heatmaps, trend reports
Deployment	Firebase / Vercel → Govt cloud	MVP fast deployment → state-scale secure rollout
ML (Future)	Python + TensorFlow/PyTorch	Auto-prioritization, duplicate detection, predictive insights

Scalable, Secure, Citizen-Centric Technology Foundation